INCH-POUND
A-A-59294
September 24, 1998
SUPERSEDING
MIL-S-3785E(AT)
24 March 1998
MS53011E
6 February 1976

COMMERCIAL ITEM DESCRIPTION

STARTERS, ENGINE, ELECTRICAL: 24-VOLT DIRECT-CURRENT

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This specification covers 24-volt (V) direct-current (dc) starters for cranking nonaircraft internal-combustion engines.

2. ClASSIFICATION. Starters will be of the following type, grade, class, and duty specified below.

Type II	-	For peripheral (ring gear) engagement.
Grade A	-	Waterproof.
Class 1	-	Positive drive - solenoid.
Duty	-	Heavy.

Beneficial comments recommendations, additions, deletions and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.

AMSC N/A FSC 2920

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

3. SALIENT CHARACTERISTICS

3.1 Design, materials, and manufacturing processes. The design, materials, and manufacturing process selection shall be the prerogative of the contractor as long as all articles submitted to the Government fully meet the physical and performance characteristics of this CID. The use of recovered material made in compliance with regulatory requirements is acceptable providing all requirements of this CID are met.

3.2 Operating characteristics.

3.2.1 Cranking torque and speed. Starter shall evidence no damage and shall conform to figure 1 when mounted on a test stand to simulate actual mounting in use. Recommended SAE practice for ring gears or starter-driven gears dimensions and pitch circle clearances shall be used.

3.2.1.1 Stalling torque. The starter shall meet the following stalling torque requirements:

a. Volts: 3.5 (approximately)
b. Current amperes (A) (max.): 500
c. Torque (pound-foot (lb-ft)) (minimum.): 16

3.2.1.2 Solenoid current consumption. Solenoid current consumption (pull-in plus hold-in) shall not exceed 81 A.

3.2.2 Position. The starter shall operate within specified performance requirements (see 3.2.1) and without evidence of damage when the longitudinal axis is in any angular position within 45 degrees of the intended operating position.

3.2.3 Overspeed capability. When subjected to 24 V properly applied at the motor terminals, the starter shall evidence no damage after operation of 1 minute at no load.

3.2.4 Dielectric strength. When subjected to an applied voltage of 500 V at a frequency of 60 hertz (Hz), the insulation shall evidence no burning, charring, loosening or cracking (arcing), smoking, rupture or other damage.

3.2.5 Loaded static torque. When energized into a stationary ring gear, pinion and clutch dentils shall have consistent engagement without hesitation and shall provide 350 lb-ft of torque at the drive.

3.2.6 Tooth abutment load. With a maximum axial force of 38 pounds (lb), applied to a single pinion tooth at its pitch diameter, the pinion shall move to its full indexing position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

3.2.7 Overrunning torque. When tested to the following procedures, pinion and clutch dentils shall be separated between 10 000 and 14 000 revolutions per minute (rpm). When pinion and clutch dentils are separated, the overrunning torque shall be less than 8 pound-inch (lb-in.) at 15 000 rpm.

a. Rotate positork drive pinion to overrunning direction by mechanical means (see figure 1).
b. Measure speed of the drive pinion at which pinion and clutch dentils separate. Separation of the dentils can be determined by change in sound.
c. Increase speed of the drive pinion to 15 000 rpm, and measure overrunning torque of the drive.

3.3 Design interface.

3.3.1 Drive. The drive shall consist of a pinion for meshing with the flywheel ring gear and a means of effecting such meshing. The pinion shall employ a positive drive for meshing with the flywheel ring gear. The pinion shall be actuated into mesh by a solenoid.

3.3.2 Mounting flange. The mounting flange shall conform to figure 2 and figure 3, and shall be adjustable to all positions (A thru L) shown in figure 3.

3.3.3 Pinion. The starter shall include one of three different pinion configurations, as applicable. The starter/pinion combinations, identified by Part or Identifying Numbers (PINs), shall conform to figure 4 and the following (see 7.1):

a. PINs AA59294-1 and AA59294-3 - 11 teeth (12 tooth blank); 6/8 pitch; 2.000 inch (in.) PD; 2.250 in. OD; 20° pressure angle; 0.015 to 0.025 in. PL clearance.
b. PIN AA59294-2 - 12 teeth (13 tooth blank); 8/10 pitch; 1.625 in. PD; 1.825 in. OD; 20° pressure angle; 0.015 to 0.025 in. PL clearance.
c. PIN AA59294-4 - 13 teeth (14 tooth blank); 8/10 pitch; 1.750 in. PD; 1.950 in. OD; 20° pressure angle; 0.015 to 0.025 in. PL clearance.

3.3.4 Terminals. Terminal location, thread size, and length shall conform to figure 4 and the following:

a. Battery - 1/2-13 UNC 2A; 0.50 in. minimum, 1.00 in. maximum length beyond outside nut when jam nut, washer and outside nut are assembled on to the stud.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

b.	Motor	- Ground, 1/2-13 UNC 2A, 0.50 maximum length beyond outside nut when starter is assembled.
c.	Switch	- No. 10-32 UNF 2A (PINs AA59294-1, 2, and 4) - 1/4-20 UNF 2A (PIN AA59294-3)

3.3.5 Rotation. Unless otherwise specified (see 7.3), the direction of the rotation of the drive, as viewed from the driving end of the starter, shall be clockwise.

3.4 Ownership and support.

3.4.1 Life test (live engine test). The starter drive shall have a minimum life of 45 000 cycles.

3.4.2 Endurance. The starter shall be sufficiently durable to complete 1000 cycles of endurance cycles on a test stand without servicing or replacement of parts. Each cycle shall consist of 30 seconds of operation under load followed by 5.5 minutes off time. The load shall be so adjusted that the starter produces 95% of the maximum horsepower shown in figure 5. The load shall be applied at the starter pinion through ring gear engagement means as in intended operation. An air velocity of approximately 1500 linear feet per minute to force air over and around the starter is allowable. During the last 200 cycles, the longitudinal axis of the starter shall be positioned at an angle with the starter pinion 45 degrees above and below the horizontal for 100 cycles each. Following the endurance test, the starter shall operate within 10 percent (%) of the performance requirements specified in 3.2.

3.4.2.1 Positive drive solenoid. The solenoid, its components, and engaging linkages shall be sufficiently durable to complete 1000 hours of endurance cycles without malfunction. The drive pinion shall not mill the flywheel or mating gear used.

3.4.2.2 Brush wear. Brushes shall be sufficiently durable to complete 1000 cycles of endurance cycling without loss of more than 50% of the original effective brush length on any one brush.

3.4.2.3 Extreme temperature. The starter and its drive and clutch assembly shall operate satisfactorily at ambient temperatures from -65 to +160 degrees Fahrenheit (°F) (-54 to 71 degrees Celsius (°C)).

3.4.3 Painting. Unless otherwise specified (see 7.3), exterior surfaces of the starter shall be painted with black gloss enamel.

3.4.4 Identification and marking. Identification and marking of the starter shall be permanent and legible and shall include the following minimum information.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

a. Starter, 24 Vdc.
b. Rotation (indicated by arrow).
c. CID A-A-59294 and Part Number (see 7.1).
d. National Stock Number (NSN).
e. Manufacturer's serial number.
f. Manufacturer's name, identification (CAGE), or Logo.

3.5 Environmental.

3.5.1 Waterproofness. The starter shall be capable of operating submerged with no accumulation of water in its interior. To determine conformance to the characteristics, the starter with its electrical connections and drive end sealed shall be submerged in a 5% saline solution with the uppermost surface a minimum of 1 in. below the surface of the solution and installed in the chamber. The motor shall be subjected to 6 psi internal vacuum for 30 minutes. While submerged, the motor shall be operated three times for 30 seconds each at 5 minutes intervals and reduced voltage of 18 to 20 V. There shall be no air leakage from the interior of the motor or water accumulation inside the motor. Air bubbles which are the results of air trapped in pockets on the exterior of the motor shall not be considered a leak.

3.5.2 Corrosion. The starter shall evidence no corrosion affecting operation when subjected to a 5% sodium chloride salt spray duration for 96 hours, after which the exposure shall be continued with the drive end exposed for 96 hours. Subsequently, the starter with its components shall meet the operating and waterproofness requirements of 3.2 and 3.5.1.

3.5.3 Fungus. The starter shall evidence no microbial growth when subjected to the fungus test specified in ASTM G21, except the test duration shall be a continuous 90 days.

3.5.4 Vibration. The starter shall function as intended with no loosened or damaged parts when subjected to a simple harmonic motion having an amplitude of 0.03 in. (0.06 in. maximum total excursion), at a frequency ranging between 10 and 55 Hz. The entire frequency range, from 10 to 55 Hz and return to 10 Hz shall be traversed in approximately 1 minute. The motion shall be applied for a period of 1 hour in each of 3 mutually perpendicular directions (total of 3 hours).

3.5.5 Shock. The starter shall function as intended with no loosened or damaged parts when exposed to severe shock conditions in accordance with figure 6.

4. REGULATORY REQUIREMENTS

4.1 Recovered material. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

5. QUALITY ASSURANCE PROVISIONS

5.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

5.2 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

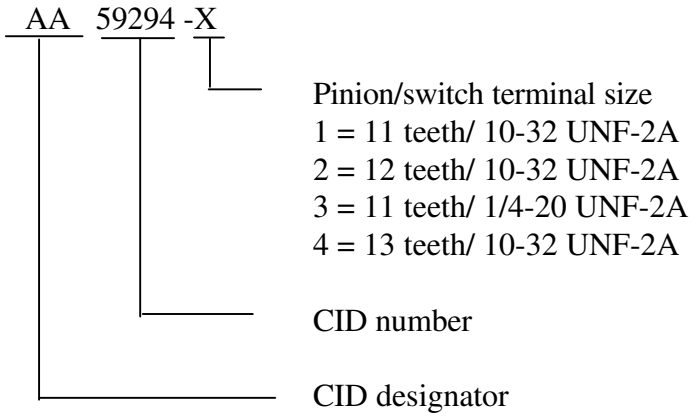
6. PACKAGING.

Preservation, packing, and marking shall be as specified in the contract or order (see 7.3).

7. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

7.1 Part or identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor. The PINs to be used for starters acquired to this CID are created as follows:



7.2 Addresses for obtaining copies of referenced documents.

7.2.1 Non-Government publication. Copies of ASTM G21 "Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi" is available from the American Society for Testing and Materials (ASTM), 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

7.3 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Direction for rotation, if other than specified (see 3.3.4).
c. Painting, if other than specified (see 3.4.3).
d. Packaging requirements (see 5.1).

7.4 Cross-reference. The starter conforming to this CID is interchangeable/substitutable with type II, grade A, class 1, heavy-duty starter conforming to MIL-S-3785E(AT), dated 24 March 1989 and MS53011, dated 6 February 1976.

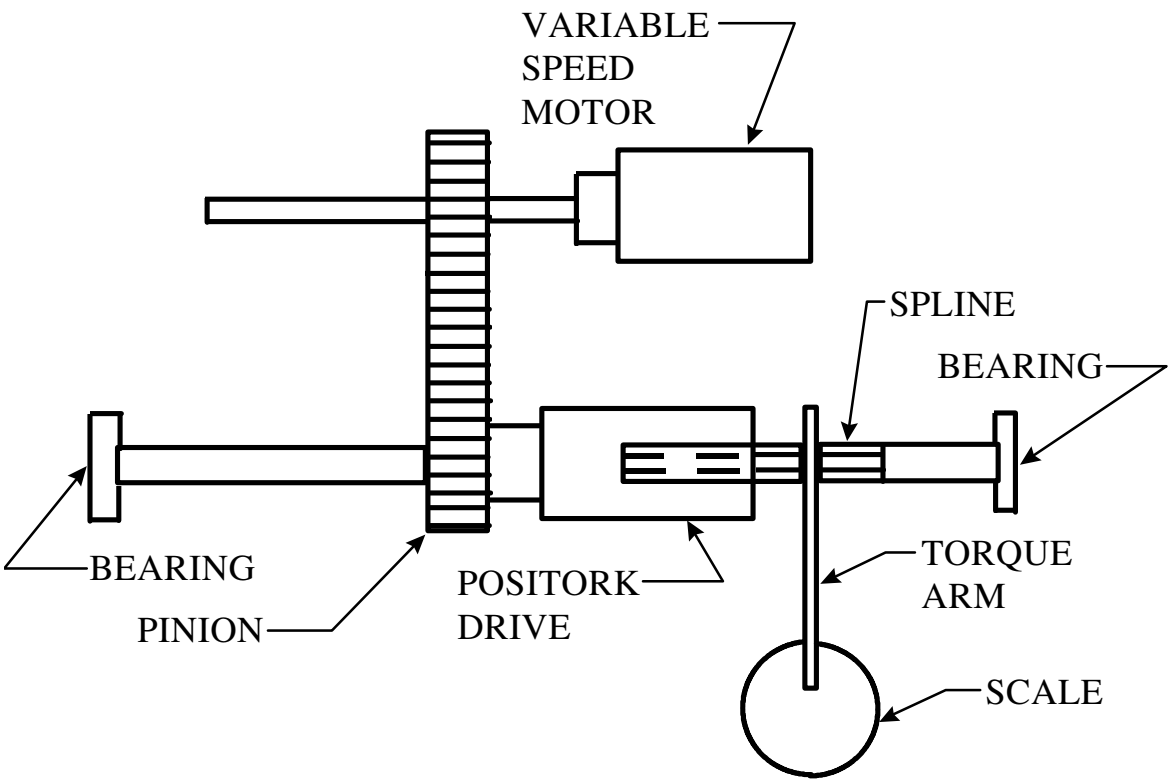


FIGURE 1. Mechanical means for conducting the overrunning torque test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

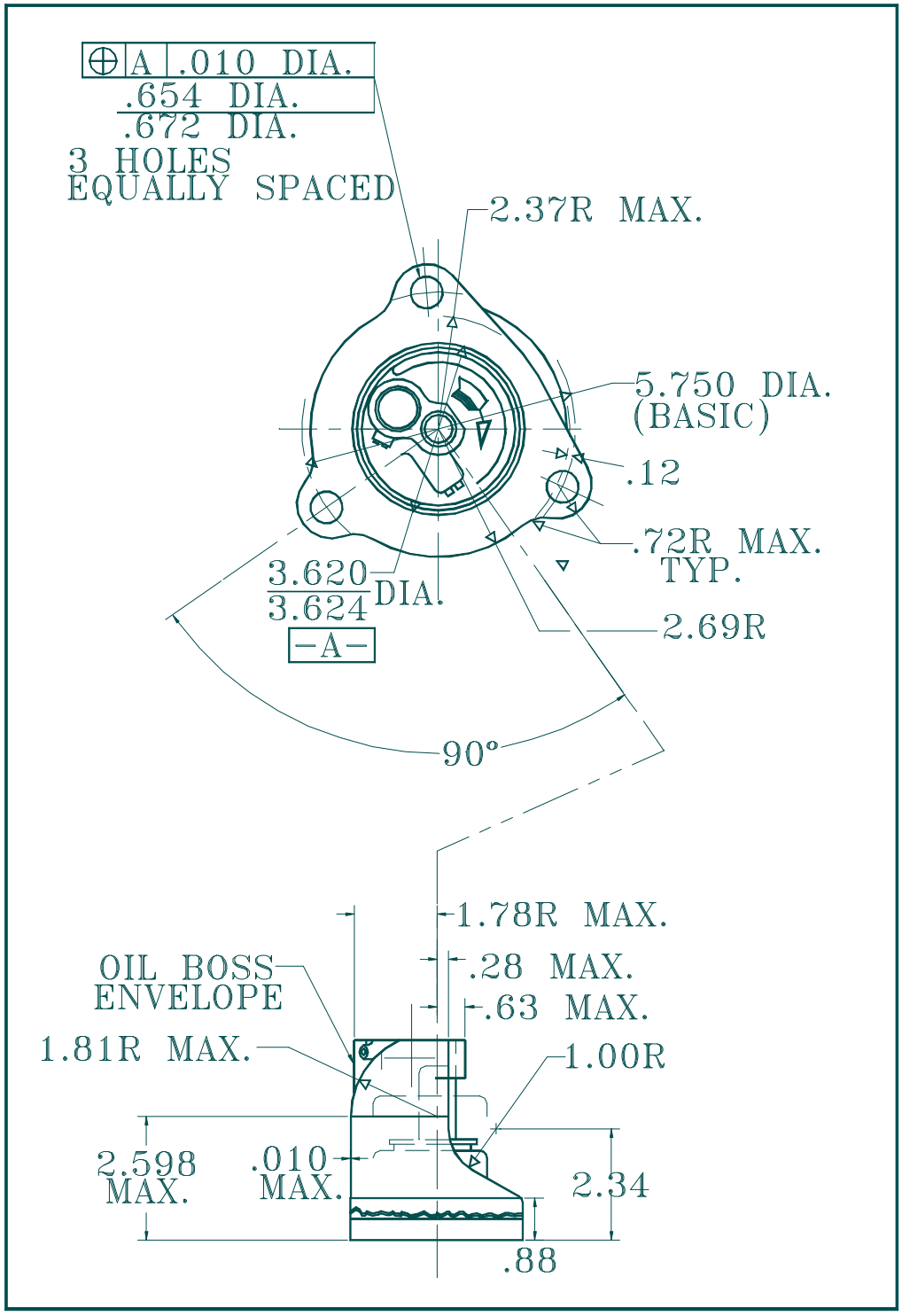


FIGURE 2. Mounting flange mean unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

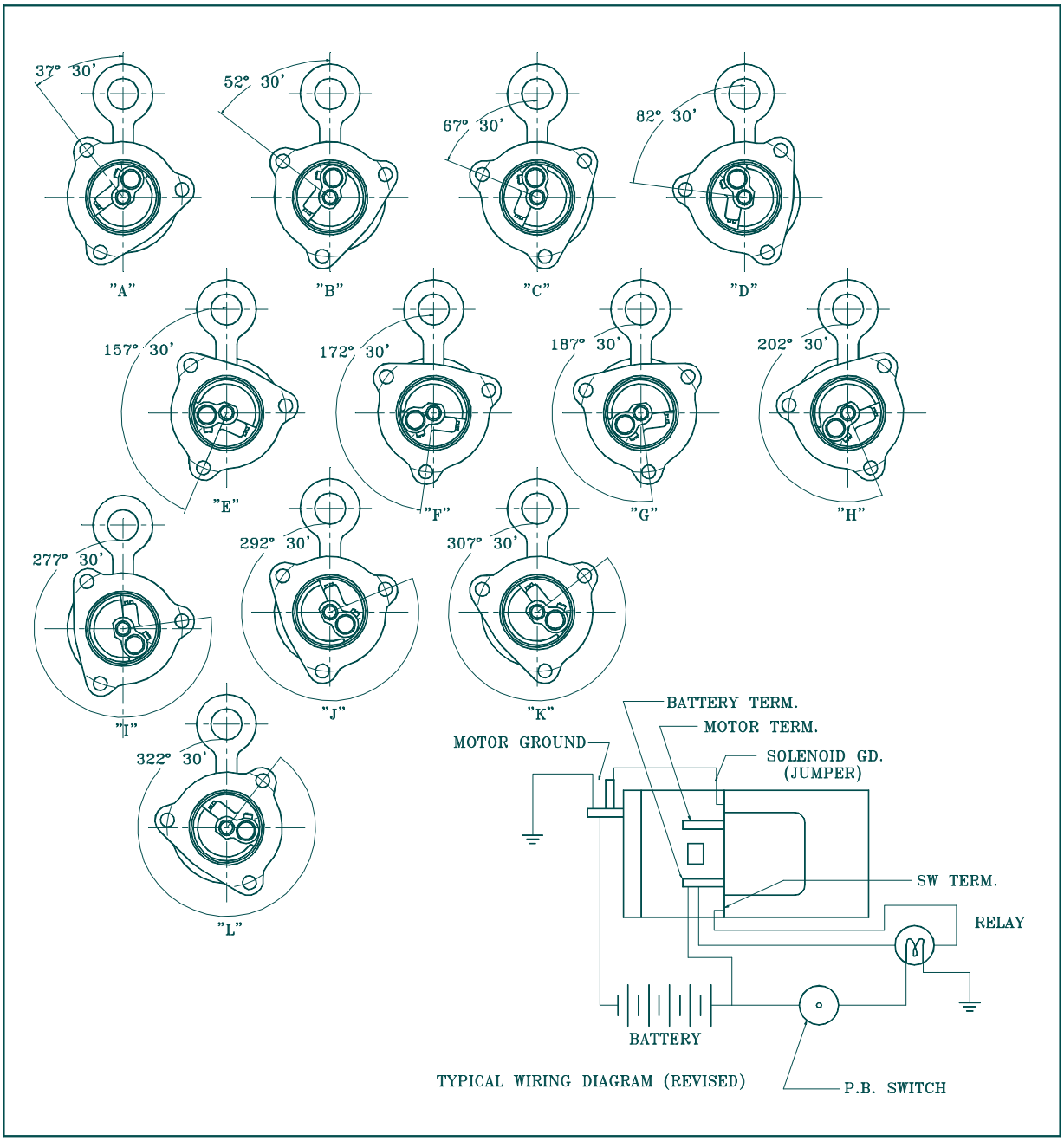
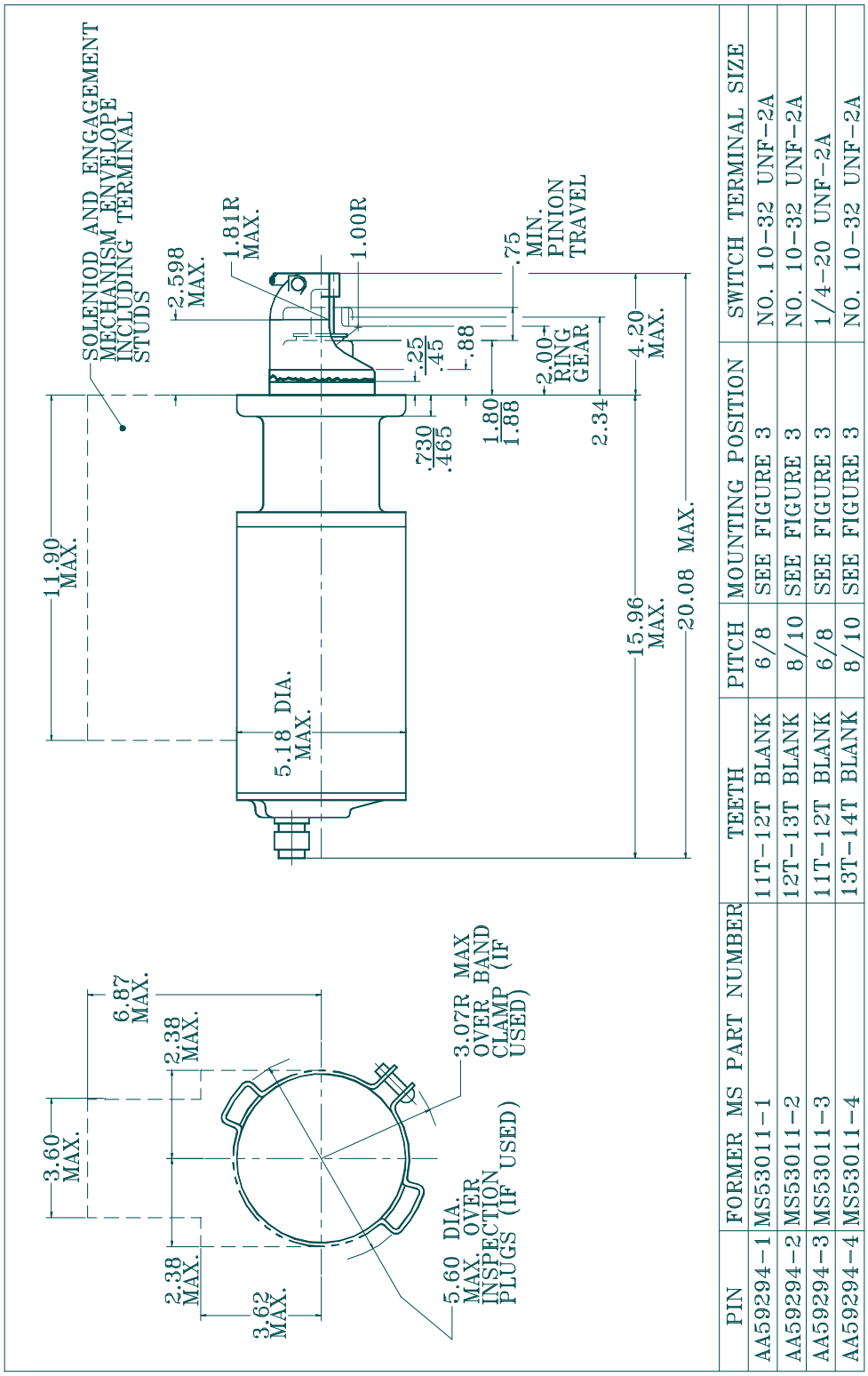


FIGURE 3. Mounting flange positions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.



PIN	FORMER MS PART NUMBER	TEETH	PITCH	MOUNTING POSITION	SWITCH TERMINAL SIZE
AA59294-1	MS53011-1	11T-12T BLANK	6/8	SEE FIGURE 3	NO. 10-32 UNF-2A
AA59294-2	MS53011-2	12T-13T BLANK	8/10	SEE FIGURE 3	NO. 10-32 UNF-2A
AA59294-3	MS53011-3	11T-12T BLANK	6/8	SEE FIGURE 3	1/4-20 UNF-2A
AA59294-4	MS53011-4	13T-14T BLANK	8/10	SEE FIGURE 3	NO. 10-32 UNF-2A

FIGURE 4. Terminal size and length

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.

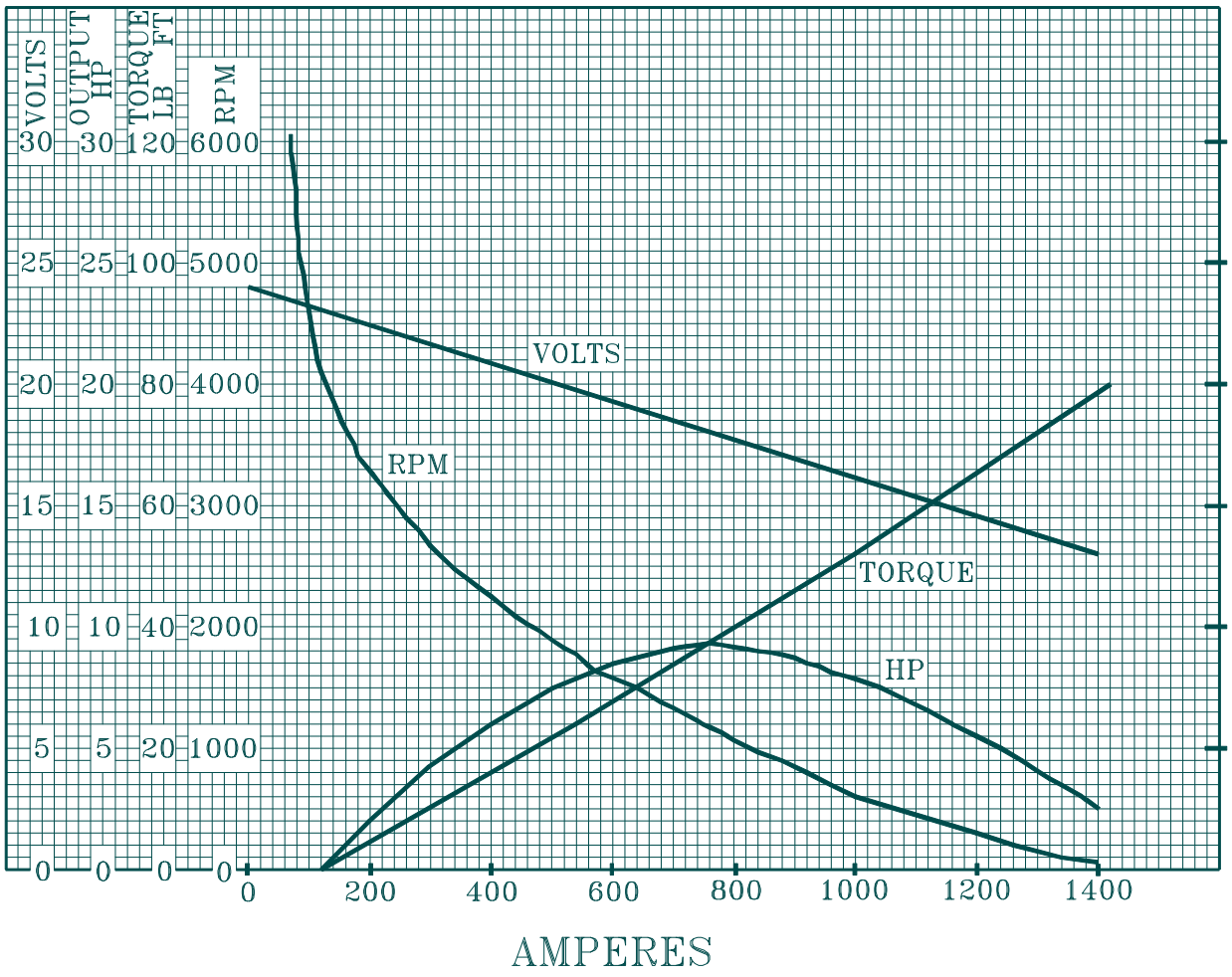
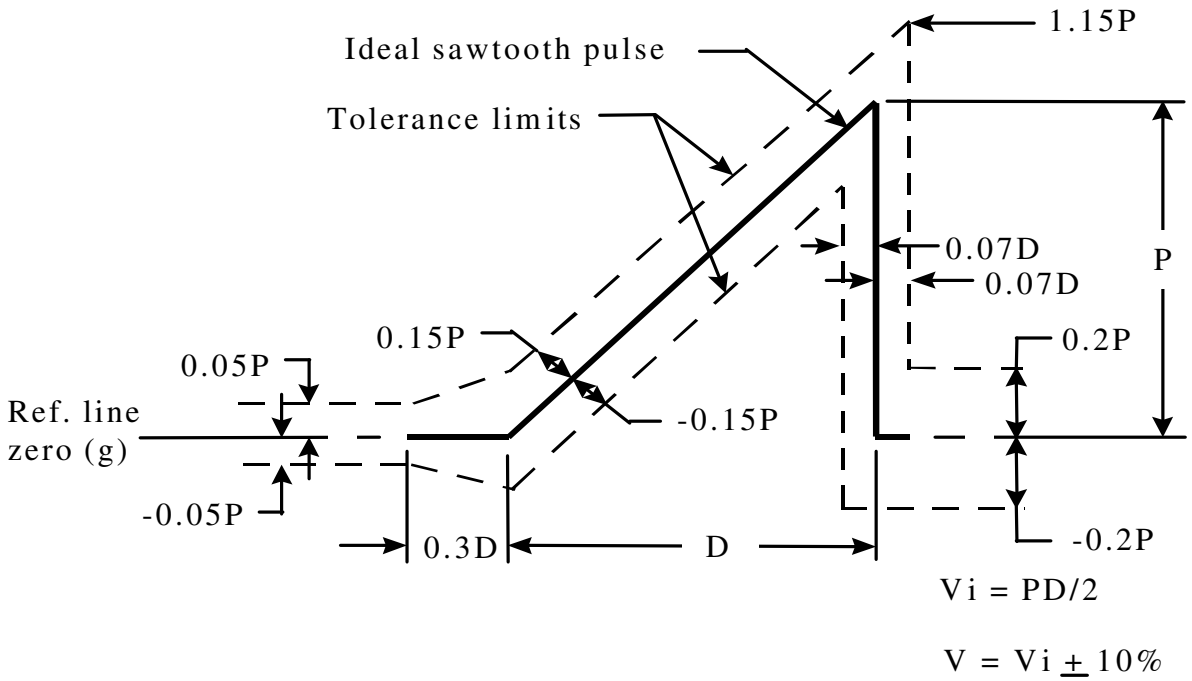


FIGURE 5. Performance curve.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.



NOTES:

1. The oscillogram should include a time about 3D long with the pulse approximately in the center. The integration to determine the velocity change should extend from 0.4D before the pulse to 0.1D beyond the pulse. The peak acceleration magnitude of the sawtooth pulse is P and its duration is D. Any measured acceleration pulse which can be contained between the broken line boundaries is a nominal terminal-peak sawtooth pulse of nominal peak value, P, and nominal duration, D. The velocity-change associated with the measured acceleration pulse is V_i.

2. P = 30(g's), D = 11 (ms), and V_i = 5.3 ft/sec.

FIGURE 6. Tolerances for terminal-peak sawtooth shock pulse.

MILITARY INTERESTS:

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FSS

Custodian:
Army - AT

Preparing Activity:
Army - AT

Review Activities:
Army - CR4
DLA - CC

(Project 2920-0497)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:09Z
Check the source to verify that this is the current version before use.